

```

37 • SELECT d.name AS Department,
38       p.name AS Head_Physician
39 FROM department d
40 JOIN physician p
41 ON d.head = p.employeeid;
42
43 SELECT *
44 FROM physician
45 WHERE position = 'Surgical Attending Physician';
46
47 SELECT *
48 FROM physician
49 WHERE name LIKE 'John%';
50
51 • SELECT COUNT(*) AS Total_Physicians
52 FROM physician;
53

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
employeeid	name	position	ssn	
3	Christopher Turk	Surgical Attending Physician	333333333	
6	Todd Quinlan	Surgical Attending Physician	666666666	
7	John Wen	Surgical Attending Physician	777777777	
* NULL	NULL	NULL	NULL	

physician 3 ×

Output			
#	Time	Action	Message
3	15:12:00	DROP TABLE IF EXISTS department	0 row(s) affected
4	15:12:03	DROP TABLE IF EXISTS physician	0 row(s) affected
5	15:12:06	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	0 row(s) affected
6	15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
8	15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
61			
62 •		SELECT position,	
63		COUNT(*) AS Total_Employees	
64		FROM physician	
65		GROUP BY position	
66		HAVING COUNT(*) > 1;	
67			
68		SELECT *	
69		FROM physician	
70		ORDER BY name ASC;	
71			

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
position	Total_Employees		
Surgical Attending Physician	3		

Result 5 ×

Output			
#	Time	Action	Message
5	15:12:06	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	0 row(s) affected
6	15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
8	15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
10	15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
11	15:15:45	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	2 row(s) returned
12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

```

122
123 • SELECT *
124 FROM department d
125 WHERE EXISTS (

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: |

	employeeid	name	position	ssn
▶	5	Bob Kelso	Head Chief of Medicine	555555555
	3	Christopher Turk	Surgical Attending Physician	333333333
	2	Elliot Reid	Attending Physician	222222222
	1	John Dorian	Staff Internist	111111111
	7	John Wen	Surgical Attending Physician	777777777

physician 6 x

Output

Action Output

#	Time	Action	Message
✓	6 15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓	7 15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	8 15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	9 15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	10 15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	11 15:15:45	SELECT * FROM physician WHERE name LIKE 'John%': LIMIT 0, 1000	2 row(s) returned
✓	12 15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓	13 15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

```

123 • SELECT *
124 FROM department d
125 WHERE EXISTS (
126     SELECT 1
127     FROM physician p
128     WHERE p.employeeid = d.head

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: |

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
•	NULL	NULL	NULL

department 7 x

Output

Action Output

#	Time	Action	Message
✓	7 15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	8 15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	9 15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
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✓	12 15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓	13 15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓	14 15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid...	3 row(s) returned

123 • **SELECT ***
 124 **FROM** department d
 125 **WHERE EXISTS** (
 126 **SELECT 1**
 127 **FROM** physician p
 128 **WHERE** p.employeeid = d.head

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: | **Result Grid**

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
4	OB/GYN	10
5	Pediatrics	11
6	Geriatrics	12
7	Neurology	13
8	Orthopedics	14
9	Cardiology	15
10	Oncology	16
11	Endocrinology	17
12	Radiology	18
13	Pathology	19
14	Pharmacy	20
15	Emergency	21
16	Intensive Care	22
17	Transplant	23
18	Neonatology	24
19	Plastic Surgery	25
20	Urology	26
21	Gynecology	27
22	Ophthalmology	28
23	Otorhinolaryngology	29
24	Dermatology	30
25	Podiatry	31
26	Cardiothoracic Surgery	32
27	Vascular Surgery	33
28	Transplant Surgery	34
29	Neurosurgery	35
30	Orthopedic Surgery	36
31	Plastic and Reconstructive Surgery	37
32	Urologic Surgery	38
33	Gynecologic Surgery	39
34	Ophthalmic Surgery	40
35	Otorhinolaryngologic Surgery	41
36	Dermatologic Surgery	42
37	Podiatric Surgery	43
38	Cardiothoracic and Vascular Surgery	44
39	Transplant and Organ Recovery Surgery	45
40	Neurological Surgery	46
41	Orthopedic and Spinal Surgery	47
42	Plastic and Reconstructive Surgery	48
43	Urologic and Urologic Surgery	49
44	Gynecologic and Gynecologic Surgery	50
45	Ophthalmic and Ophthalmic Surgery	51
46	Otorhinolaryngologic and Otorhinolaryngologic Surgery	52
47	Dermatologic and Dermatologic Surgery	53
48	Podiatric and Podiatric Surgery	54
49	Cardiothoracic and Vascular Surgery	55
50	Transplant and Organ Recovery Surgery	56
51	Neurological Surgery	57
52	Orthopedic and Spinal Surgery	58
53	Plastic and Reconstructive Surgery	59
54	Urologic and Urologic Surgery	60
55	Gynecologic and Gynecologic Surgery	61
56	Ophthalmic and Ophthalmic Surgery	62
57	Otorhinolaryngologic and Otorhinolaryngologic Surgery	63
58	Dermatologic and Dermatologic Surgery	64
59	Podiatric and Podiatric Surgery	65
60	Cardiothoracic and Vascular Surgery	66
61	Transplant and Organ Recovery Surgery	67
62	Neurological Surgery	68
63	Orthopedic and Spinal Surgery	69
64	Plastic and Reconstructive Surgery	70
65	Urologic and Urologic Surgery	71
66	Gynecologic and Gynecologic Surgery	72
67	Ophthalmic and Ophthalmic Surgery	73
68	Otorhinolaryngologic and Otorhinolaryngologic Surgery	74
69	Dermatologic and Dermatologic Surgery	75
70	Podiatric and Podiatric Surgery	76
71	Cardiothoracic and Vascular Surgery	77
72	Transplant and Organ Recovery Surgery	78
73	Neurological Surgery	79
74	Orthopedic and Spinal Surgery	80
75	Plastic and Reconstructive Surgery	81
76	Urologic and Urologic Surgery	82
77	Gynecologic and Gynecologic Surgery	83
78	Ophthalmic and Ophthalmic Surgery	84
79	Otorhinolaryngologic and Otorhinolaryngologic Surgery	85
80	Dermatologic and Dermatologic Surgery	86
81	Podiatric and Podiatric Surgery	87
82	Cardiothoracic and Vascular Surgery	88
83	Transplant and Organ Recovery Surgery	89
84	Neurological Surgery	90
85	Orthopedic and Spinal Surgery	91
86	Plastic and Reconstructive Surgery	92
87	Urologic and Urologic Surgery	93
88	Gynecologic and Gynecologic Surgery	94
89	Ophthalmic and Ophthalmic Surgery	95
90	Otorhinolaryngologic and Otorhinolaryngologic Surgery	96
91	Dermatologic and Dermatologic Surgery	97
92	Podiatric and Podiatric Surgery	98
93	Cardiothoracic and Vascular Surgery	99
94	Transplant and Organ Recovery Surgery	100

department 7 x Apply

Output

Action Output

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7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
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12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1	1 row(s) returned
13	15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.head)	3 row(s) returned

131 • **SELECT MAX(employeeid) AS Maximum_Employee_ID**
 132 **FROM** physician;
 133
 134 • **SELECT MIN(employeeid) AS Minimum_Employee_ID**

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: | **Result Grid**

Minimum_Employee_ID
1

Result 10 x

Output

Action Output

#	Time	Action	Message
10	15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
11	15:15:45	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	2 row(s) returned
12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1	1 row(s) returned
13	15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.head)	3 row(s) returned
15	15:17:19	SELECT MAX(employeeid) AS Maximum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
16	15:17:42	SELECT MAX(employeeid) AS Maximum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
17	15:19:00	SELECT MIN(employeeid) AS Minimum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

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